

An Exact Counterexample to a Modulo-Five Congruence for $\bar{b}_{4,4}^2$

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Abstract

Let

$$\sum_{N \geq 0} \bar{b}_{4,4}^2(N) q^N = \frac{(q^2; q^2)_\infty^8}{(q; q)_\infty^{16}}.$$

We disprove the assertion $\bar{b}_{4,4}^2(5n) \equiv 0 \pmod{5}$ for every $n \geq 1$. Formal logarithmic differentiation gives an exact recurrence for the coefficients, from which $\bar{b}_{4,4}^2(5) = 25056 \equiv 1 \pmod{5}$. Thus the least admissible value, $n = 1$, is already a counterexample. The closely related conjecture in the motivating source has tuple parameter $k = 4$, not $k = 2$, and is not addressed by this counterexample.

1 Statement and parameter scope

For a formal variable q and a positive integer d , write

$$f_d := (q^d; q^d)_\infty := \prod_{m \geq 1} (1 - q^{dm}).$$

The generating function introduced by Thejitha and Fathima is

$$\sum_{N \geq 0} \bar{b}_{r,s}^k(N) q^N = \frac{f_2^{(3s-2r)k}}{f_1^{2sk} f_4^{(s-r)k}}. \quad (1)$$

See [1, Eq. (1.2), version 1]. Substitution of $(r, s, k) = (4, 4, 2)$ in (1) gives the formal power series

$$F(q) := \sum_{N \geq 0} b(N) q^N = \frac{f_2^8}{f_1^{16}} = \prod_{m \geq 1} (1 - q^{2m})^8 (1 - q^m)^{-16}, \quad b(N) := \bar{b}_{4,4}^2(N). \quad (2)$$

Every coefficient in this product depends on only finitely many factors, so (2) is well defined in $\mathbb{Z}[[q]]$.

The question resolved here is the universal assertion

$$b(5n) \equiv 0 \pmod{5} \quad (n \geq 1). \quad (3)$$

The source comparison requires care: Conjecture 7.1 of [1] lists $\bar{b}_{4,4}^4(5n) \equiv 0 \pmod{5}$, with $k = 4$. In contrast, (3) has $k = 2$. They are different statements.

2 The exact counterexample

Theorem 1. *For the coefficients defined by (2),*

$$b(5) = 25056 = 5 \cdot 5011 + 1.$$

Consequently, (3) is false, with counterexample $n = 1$.

Proof. All operations below take place in the formal power-series ring. Because $F(0) = 1$, formal logarithmic differentiation of (2) is legitimate and gives

$$\begin{aligned} \frac{qF'(q)}{F(q)} &= 16 \sum_{m,r \geq 1} mq^{mr} - 16 \sum_{m,r \geq 1} mq^{2mr} \\ &= \sum_{j \geq 1} c_j q^j, \end{aligned}$$

where, with $\sigma_1(j)$ denoting the sum of the positive divisors of j ,

$$c_j = \begin{cases} 16\sigma_1(j), & j \text{ odd}, \\ 16(\sigma_1(j) - \sigma_1(j/2)), & j \text{ even}. \end{cases} \quad (4)$$

Comparing the coefficient of q^N in $qF' = F \sum_{j \geq 1} c_j q^j$ yields the exact recurrence

$$Nb(N) = \sum_{j=1}^N c_j b(N-j), \quad b(0) = 1. \quad (5)$$

For $1 \leq j \leq 5$, formula (4) gives

$$(c_1, c_2, c_3, c_4, c_5) = (16, 32, 64, 64, 96).$$

Successive use of (5) therefore gives

$$(b(0), b(1), b(2), b(3), b(4), b(5)) = (1, 16, 144, 960, 5264, 25056).$$

In particular, the final recurrence step is

$$5b(5) = 16(5264) + 32(960) + 64(144) + 64(16) + 96 = 125280,$$

and hence $b(5) = 25056 \equiv 1 \pmod{5}$. Since the quantifier in (3) includes $n = 1$, this single exact coefficient disproves the universal assertion. $\square$

3 Relation to the motivating conjecture

Equation (1) shows that the $k = 4$ series appearing in the source conjecture is f_2^{16}/f_1^{32} , whereas the series tested in (2) is f_2^8/f_1^{16} . Theorem 1 therefore resolves only the stated $k = 2$ problem. It neither proves nor disproves the $k = 4$ congruence in [1, Conjecture 7.1]. The source comparison of partition congruences is limited to that directly motivating paper; no claim of an exhaustive literature search or of priority is made.

The use of generating-function coefficient analysis alongside separate exact finite diagnostics provides a methodological comparison with Zeng's study of distinct cycle lengths [2]. Here the finite calculation produces a counterexample at one index; it is not used to infer a pattern at all indices. No result on cycle lengths is needed for the partition calculation above.

Computational and AI assistance disclosure

Two Python 3 programs using exact integer arithmetic independently checked the coefficients through degree five: one multiplied the uncanceled Euler factors with sparse dictionaries, and the other compared uncanceled and canceled dense convolutions. No floating-point computation was used. These checks are supplementary; the proof above is the displayed formal recurrence and integer calculation. An OpenAI Codex system assisted with the exact computations, proof checking, source comparison, and manuscript preparation.

References

- [1] M. P. Thejitha and S. N. Fathima. *Overcolored Partition k -tuples Restricted by Parity of the Parts*. Version 1, submitted 3 September 2026. 2026. arXiv: 2609.03926v1 [math.NT]. URL: <https://arxiv.org/abs/2609.03926v1>.
- [2] Z. Zeng. “The Distinct-Cycle-Length Probability is Strictly Decreasing after 30”. 2026. DOI: 10.2139/ssrn.7380519. URL: <https://doi.org/10.2139/ssrn.7380519>.